

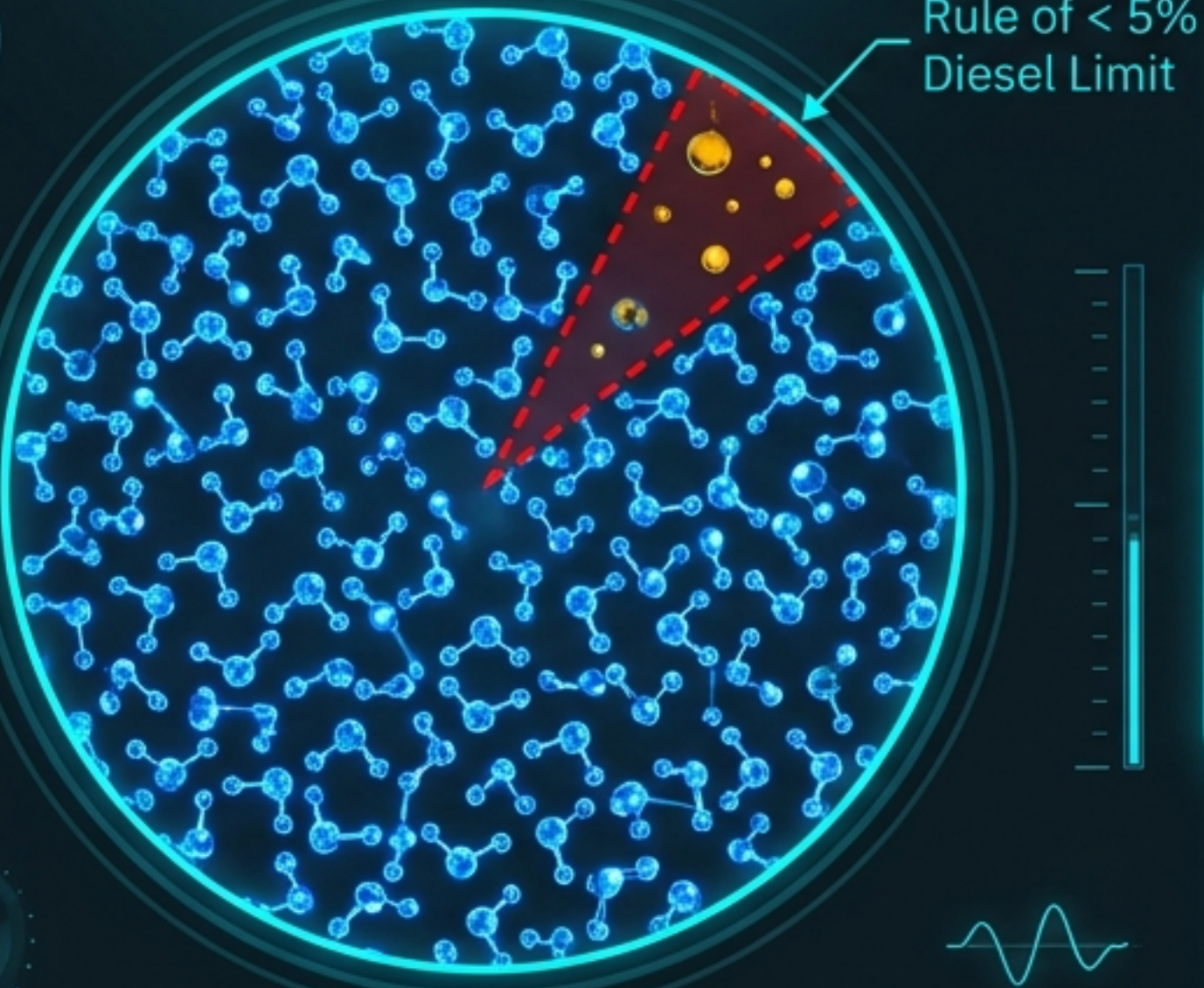


THE TRUE COST OF DRILLING FLUIDS: A COMPARATIVE ANALYSIS OF WBM AND OBM

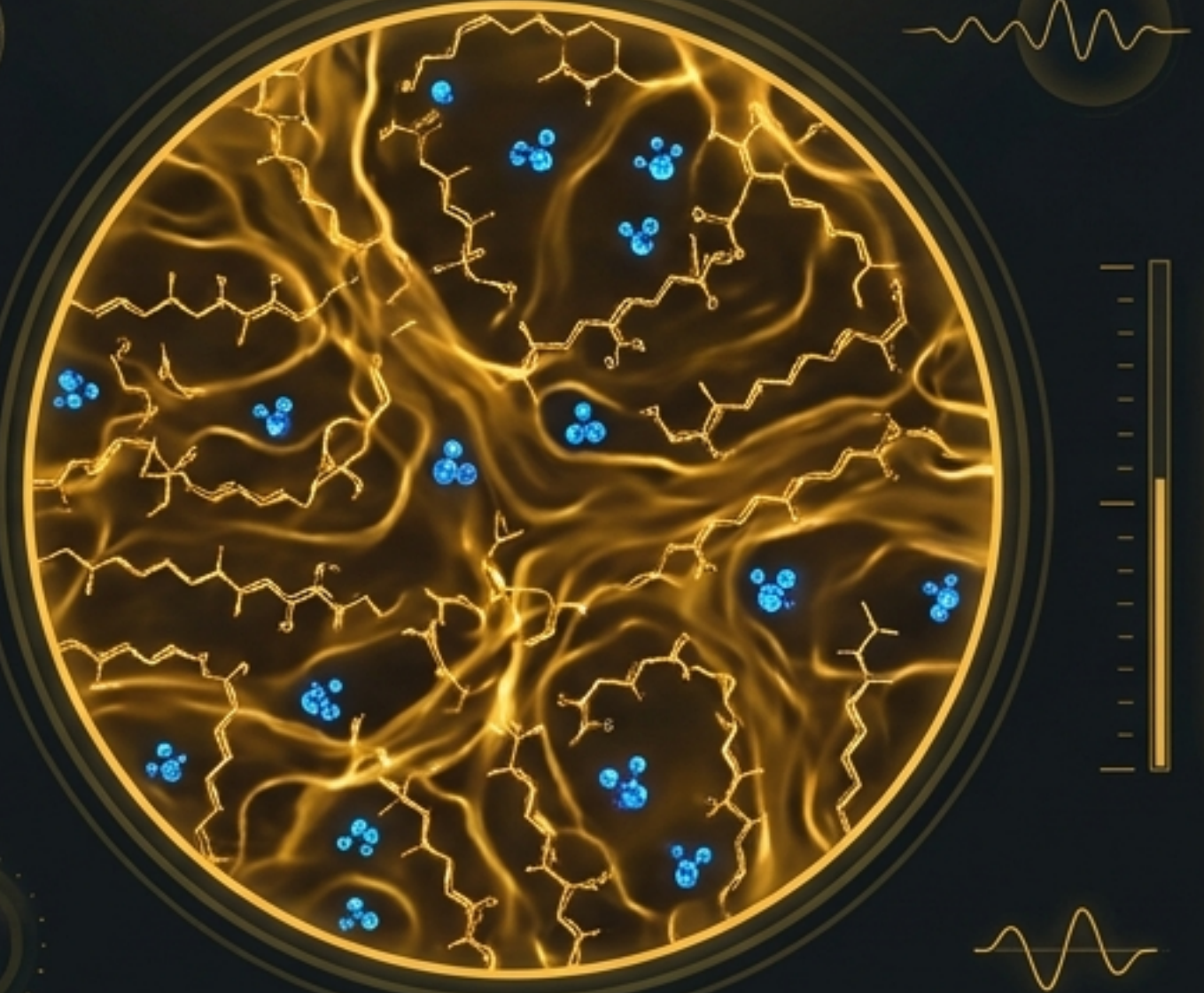
Rethinking Mud Economics, Rheology,
and Wellbore Stability

PRESENTATION BASED ON INSIGHTS BY ENGINEER SAJJAD K. HANO

Defining the Baseline: The Continuous Phase Dichotomy



WBM: Relies on water as the primary continuous phase.



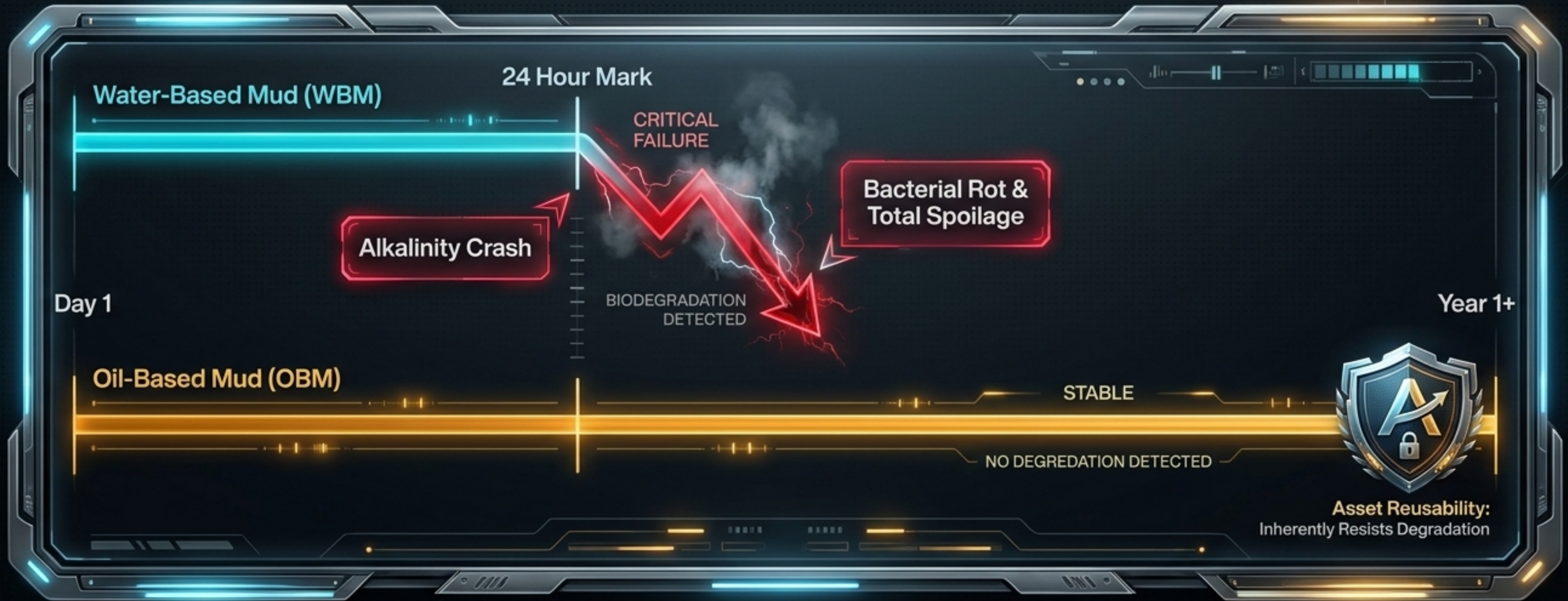
OBM: Relies on diesel or synthetic oils as the continuous phase, with minimal water.

The Cost Illusion:

Upfront Additives vs. Hidden Lifecycle Burdens

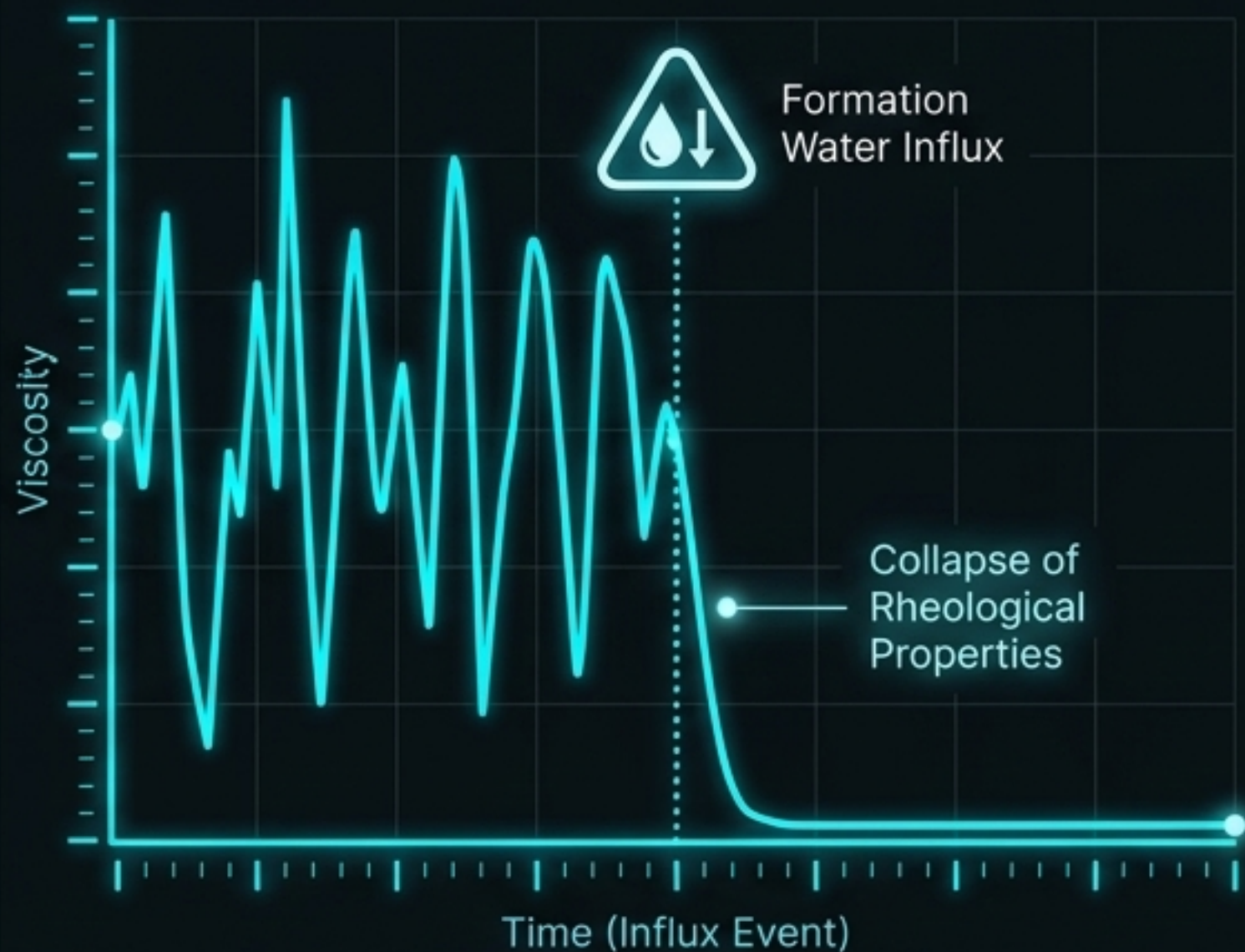


The Maintenance Burden: Biological Decay vs. Long-Term Stability

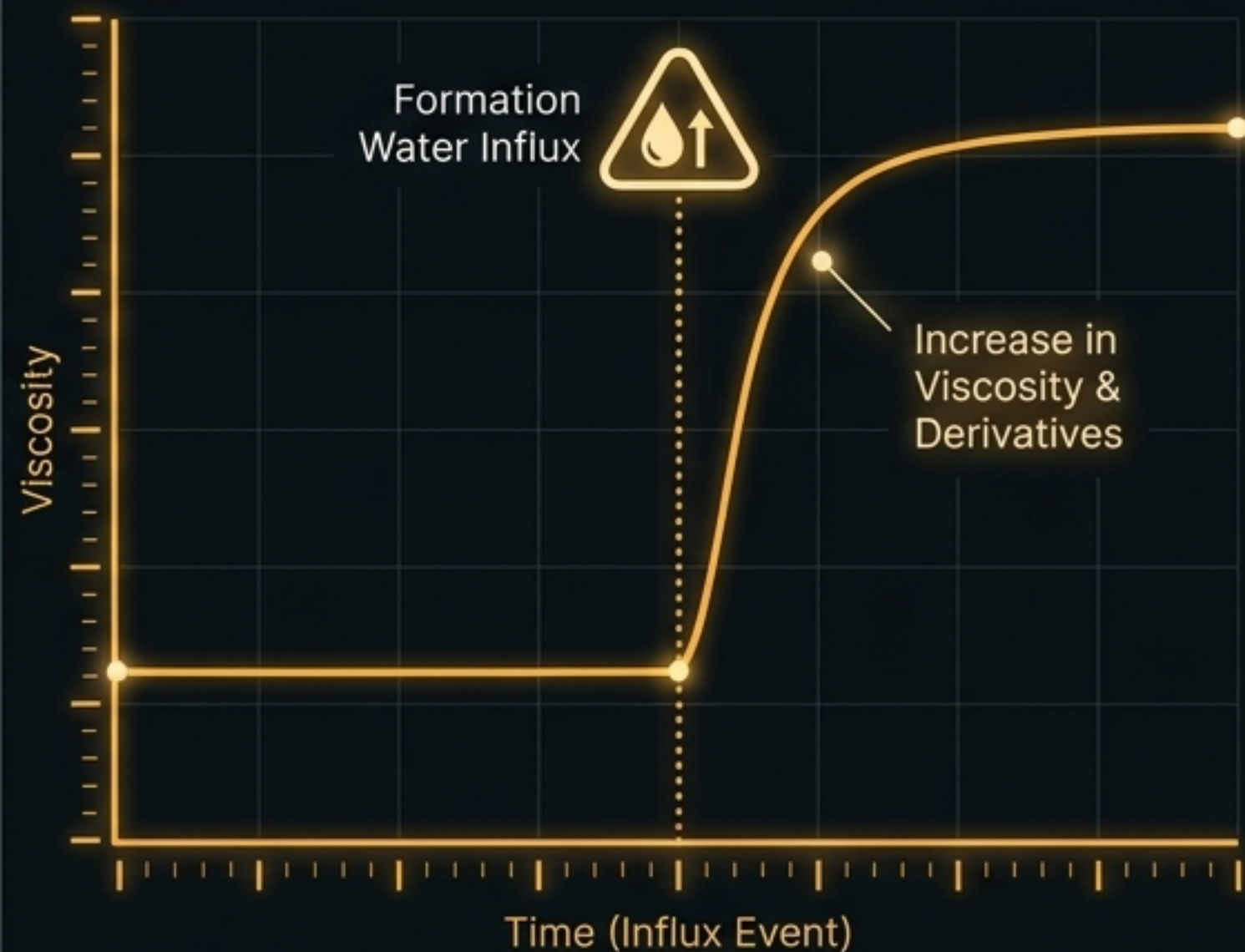


The Rheology Paradox: How Water Influx Triggers Opposite Reactions

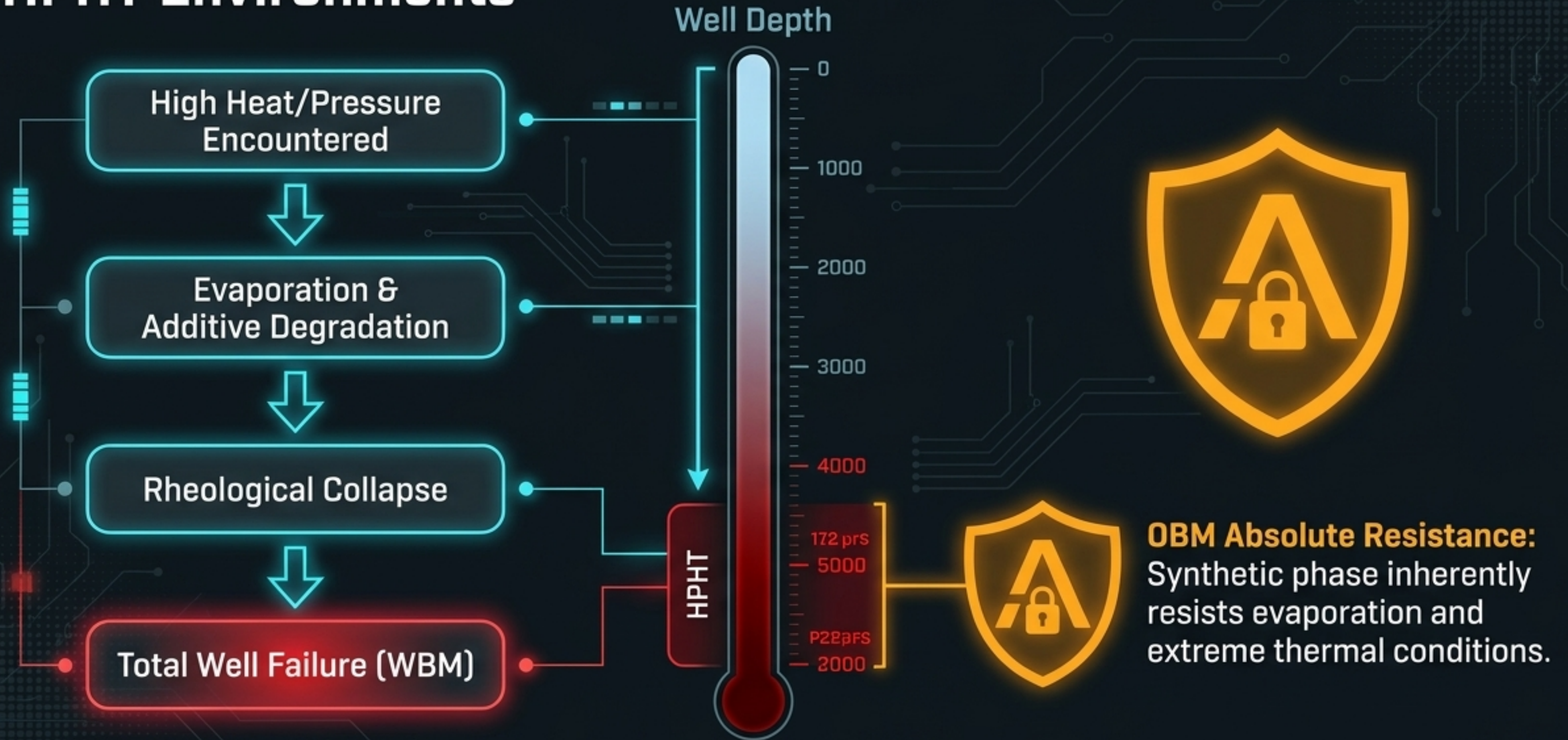
WBM: Viscosity Collapse



OBM: Viscosity Surge

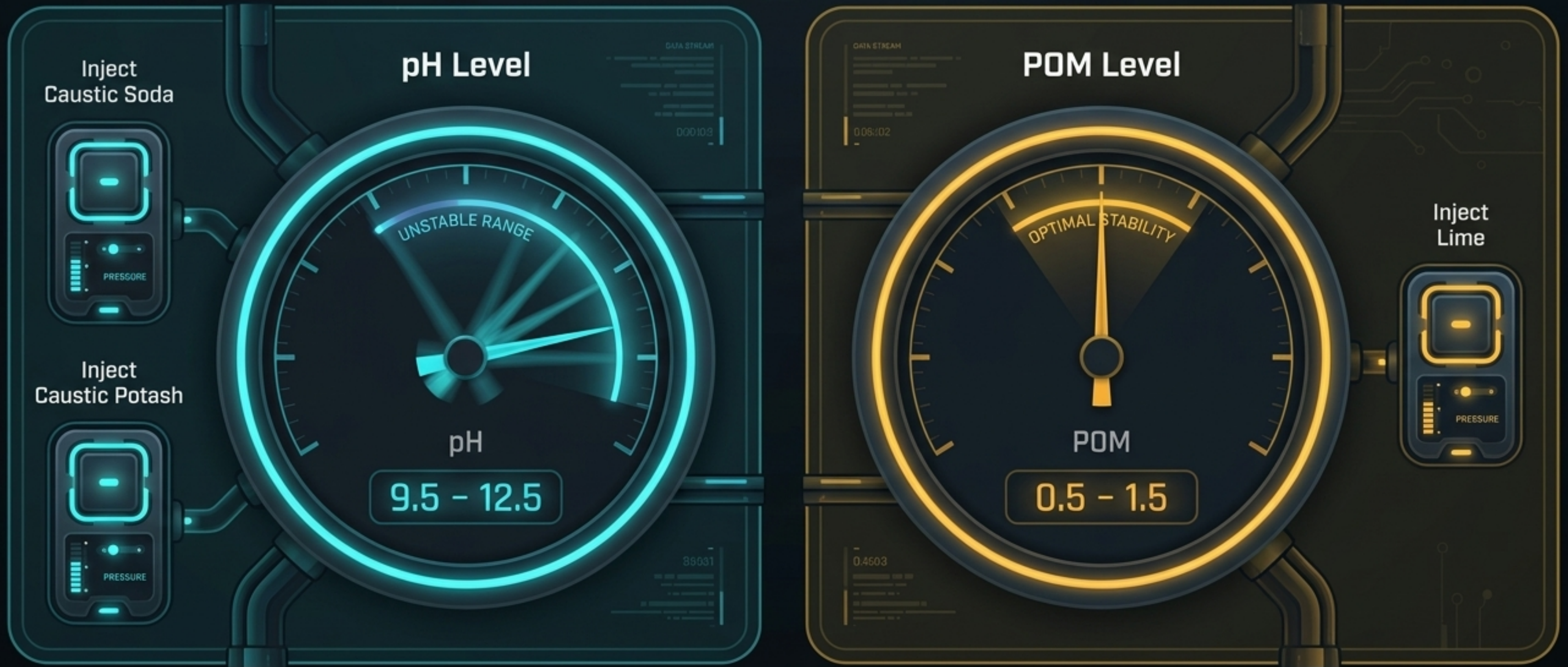


Reverse Engineering Well Failure in HPHT Environments

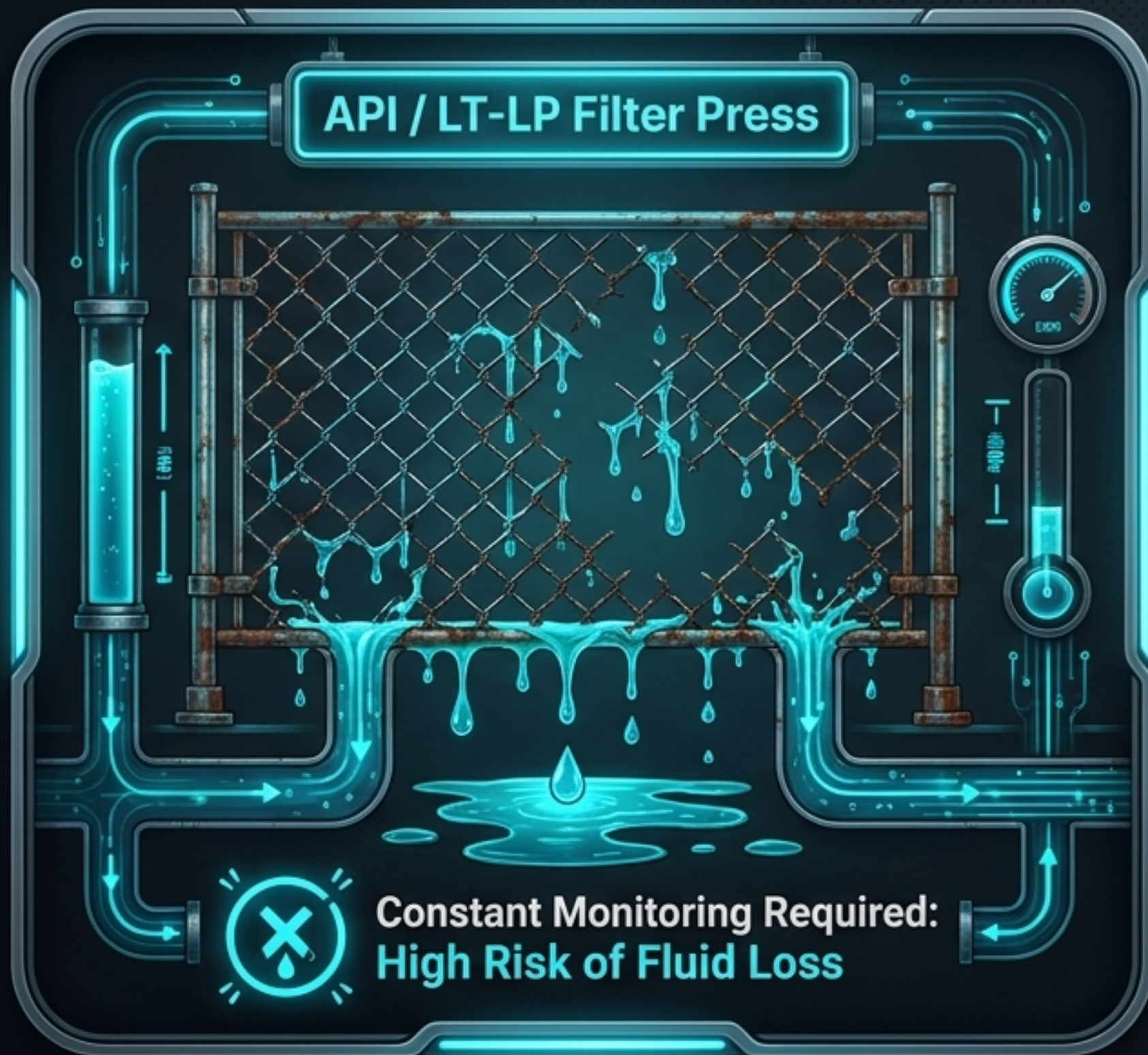


Controlling the Chemistry: Alkalinity Management Systems

Alkalinity is the lifeblood of mud maintenance, but the mechanisms of control differ fundamentally.



The Filtration Gateway: API Volatility vs. HT-HP Security



The Limits of OBM: Why Flexibility Still Demands WBM

Low Solid Polymer

KCL Polymer

Free Clay
KCL Polymer

Salt
Saturated Mud

Drill-in
Mud

Amine Mud

Amine Mud

WBM: Highly adaptable for water-sensitive formations.



OBM: Rigid, singular pattern design. Unchangeable.

The Exploratory Dilemma: Navigating the Unknown

Exploratory wells are geological black boxes. The choice of fluid dictates the survival of the well against unforeseen variables.



Expert Panel: Engineering the Exploratory Well (Part 1)



This panel features a blue-themed interface. On the left, there are two stylized female avatars. Above the top avatar is a 3D block diagram of geological strata. A central speech bubble contains the text. To the right of the speech bubble is a vertical progress bar with several segments. At the bottom left, there is a circular gauge labeled 'DEPTH' and a horizontal bar chart.

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We are hitting highly sensitive shale. WBM will cause rapid swelling and collapse. We must deploy OBM to protect protect the formation integrity.

>>>



This panel features a yellow-themed interface. On the right, there are two stylized female avatars. Above the top avatar is an icon of a gas well with a flame and smoke. A central speech bubble contains the text. To the left of the speech bubble is a vertical progress bar with several segments. At the bottom right, there is a circular gauge labeled 'MSS' and a horizontal bar chart.

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We are flying blind into gas kicks and pollutants. WBM contaminates instantly and loses rheology. OBM natively resists these pollutants, ensuring our safety margin.

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Expert Panel: The Financial Verdict (Part 2)



Project Accountant

“WBM looks cheaper on the daily spread rate. But in an exploratory well, instant contamination forces chemical additions and rig downtime that bankrupt the AFE. Here, the high upfront cost of OBM isn’t an expense—it is a mandatory insurance policy.”

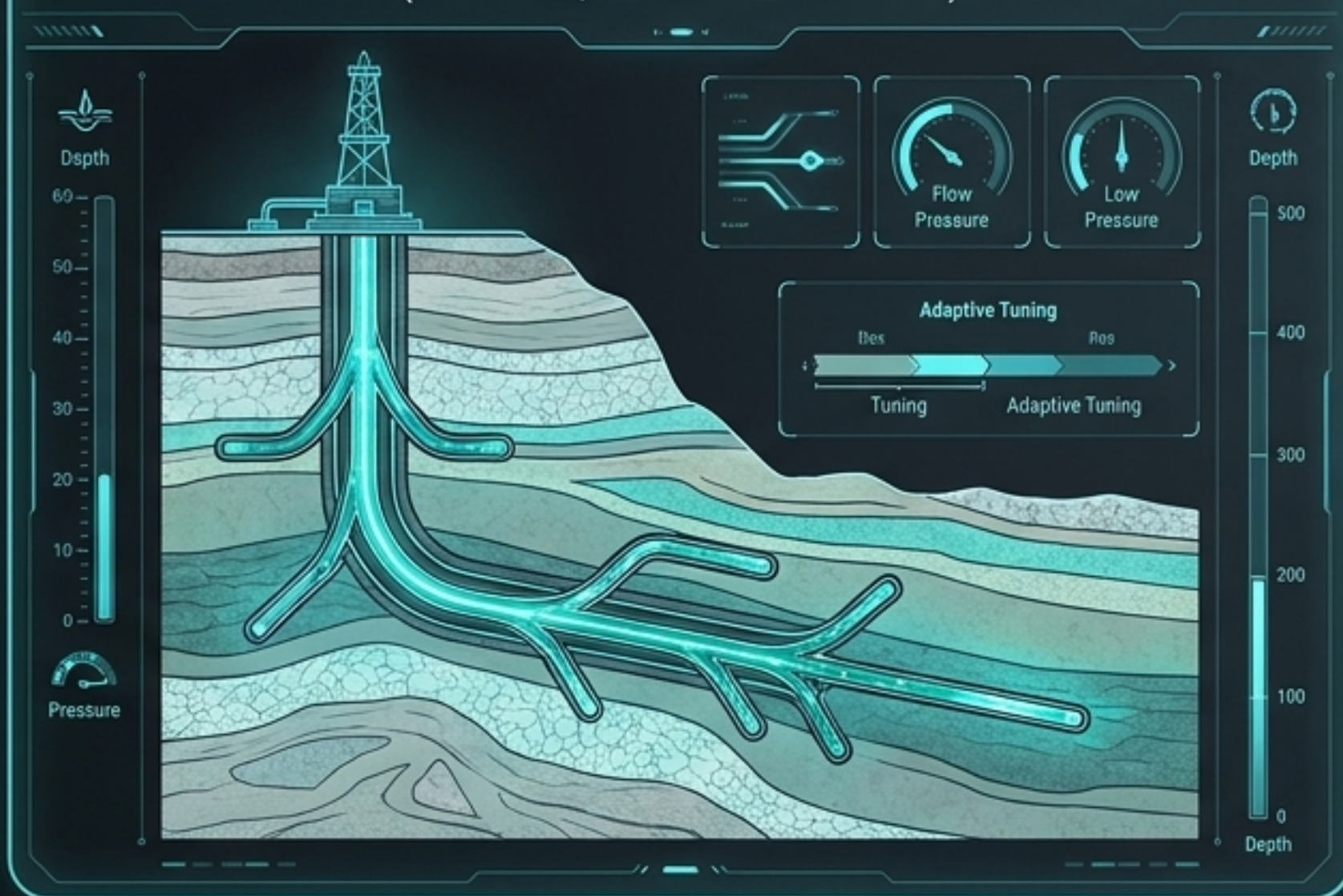
The Fluid Selection Diagnostic Matrix

Operational Dimension	WBM Profile	OBM Profile
Upfront Cost	Low	High
Lifecycle Cost (Complex Wells)	High / Unpredictable	Fixed / Low
HPHT Stability	Fails	Highly Stable
Storage & Reusability	Spoils in 24h	Years / Reusable
Rheological Volatility	High / Continuous Change	Low / Stable
Formation Adaptability	High / Custom Variants	Low / Single Pattern
Exploratory Suitability	High Contamination Risk	Preferred / Resistant

Rethinking Mud Economics: Fluid as a Structural Asset

Drilling fluid is not a consumable line item; it is a structural component of the well architecture.

WBM DEPLOYMENT: Adapted, Tuned, Economical (For known, shallow environments)



OBM DEPLOYMENT: Stable, Resilient, Assured (For depth, thermal extremes, and exploration)

